

EllipticToken

```
// SPDX-License-Identifier: MIT
pragma solidity 0.8.28;

import {Ownable} from "openzeppelin-contracts-08/access/Ownable.sol";
import {ECDSA} from "openzeppelin-contracts-08/utils/cryptography/ECDSA.sol";
import {ERC20} from "openzeppelin-contracts-08/token/ERC20/ERC20.sol";

contract EllipticToken is Ownable, ERC20 {
    error HashAlreadyUsed();
    error InvalidOwner();
    error InvalidReceiver();
    error InvalidSpender();

    constructor() ERC20("EllipticToken", "ETK") {}

    mapping(bytes32 => bool) public usedHashes;

    function redeemVoucher(
        uint256 amount,
        address receiver,
        bytes32 salt,
        bytes memory ownerSignature,
        bytes memory receiverSignature
    ) external {
        bytes32 voucherHash = keccak256(abi.encodePacked(amount, receiver,
salt));
        require(!usedHashes[voucherHash], HashAlreadyUsed());

        // Verify that the owner emitted the voucher
        require(ECDSA.recover(voucherHash, ownerSignature) == owner(),
InvalidOwner());

        // Verify that the receiver accepted the voucher
        require(ECDSA.recover(voucherHash, receiverSignature) == receiver,
InvalidReceiver());

        // Nullify the voucher
        usedHashes[voucherHash] = true;

        // Mint the tokens
        _mint(receiver, amount);
    }
}
```

```

    }

    function permit(uint256 amount, address spender, bytes memory
tokenOwnerSignature, bytes memory spenderSignature)
        external
    {
        bytes32 permitHash = keccak256(abi.encode(amount));
        require(!usedHashes[permitHash], HashAlreadyUsed());
        require(!usedHashes[bytes32(amount)], HashAlreadyUsed());

        // Recover the token owner that emitted the permit
        address tokenOwner = ECDSA.recover(bytes32(amount),
tokenOwnerSignature);

        // Verify that the spender accepted the permit
        bytes32 permitAcceptHash = keccak256(abi.encodePacked(tokenOwner,
spender, amount));
        require(ECDSA.recover(permitAcceptHash, spenderSignature) ==
spender, InvalidSpender());

        // Nullify the permit
        usedHashes[permitHash] = true;

        // Approve the spender
        _approve(tokenOwner, spender, amount);
    }
}

```

so the first thing the challenge description tells us is that Bob creates vouchers off-chain for a receiver (Alice in this case) and then Alice redeems the voucher, by looking at the challenge code we notice that the function `redeemVoucher` is the one used for the redeeming process

the challenge description tells us that our goal is to steal the tokens that Alice **just redeemed** by using the fact that Bob skipped parts of the `ECDSA` algorithm

to solve the challenge, lets start by looking in the contract for any logic that allows us to steal Alice's tokens. After analyzing the code we can notice that the only line that actually does that is `_approve(tokenOwner, spender, amount);` located at the end of the `permit` function, and this will only work if `tokenOwner` is Alice's address and `amount > 0`

so `tokenOwner` is computed in this line :

```
address tokenOwner = ECDSA.recover(bytes32(amount), tokenOwnerSignature);
```

this means in order to make it equal to Alice's address we should craft a pair amount,signature that gets verified into Alice's address

if you notice, recover is using the amount directly without hashing it, this is actually a vulnerability in *ecdsa* verification, the message should be hashed before verification, so we can say/assume that this is the missing step the challenge description talked about

for now we have 2 useful information :

- 1- we should make `tokenOwner` equal to Alice's address
- 2- the message isn't hashed, this should be the thing we will abuse

from now i will be talking about my attempts :

1- use the same hash and signature used in the redemption

we stated before that Alice just redeemed the voucher, so the function `redeemVoucher` should have been called and executed after we deployed that challenge, and since *Ethereum* is a public Blockchain, we can read the exact parameters values used in the call

ethernaut challenges are deployed in sepolia testnet, lets go and find the contract challenge there :

The screenshot shows the Etherscan interface for the address `0x6d9d949407416976aCBf4DD99535538994cc4b2b`. The page is divided into three main sections: Overview, More Info, and Multichain Info. The Overview section shows an ETH balance of 0. The More Info section shows transaction statistics (Latest: N/A, First: N/A) and funding information (Funded by: N/A). The Multichain Info section shows N/A. Below these sections, there are tabs for Transactions, Internal Transactions, and Token Transfers (ERC-20). The Transactions tab is selected, and a table with columns Transaction Hash, Method, Block, Age, From, To, Amount, and Txn Fee is displayed. The table is empty, and a message states "There are no matching entries" with a link to "5 Internal Transactions found." at the bottom.

now by our previous assumption, there should be a transaction that called the redeem function, but where is it ??

well the explanation for this is that the initial transaction that started the challenge instance is the one that called the redeem function, so the call we are looking for will be found inside an internal transaction

so we check the internal transactions :

this will match each argument with the input value

the output will be :

```
10000000000000000000 [1e19]
0xA11CE84AcB91Ac5980A4E2945C9157eF3Ab17D4e
0x04a078de06d9d2ebd86ab2ae9c2b872b26e345d33f988d6d5d875f94e9c8ee1e
0x085a4f70d03930425d3d92b19b9d4e37672a9224ee2cd68381a9854bb3673ef86b35cfdeee0fb1d2168587fb188ee46109af063bf85d9d3d6859ceb4451c
0xab1dcd2a2a1c697715a62eb6522b7999d04aa952ffa2619988737ee675d9494f2b50ecce40040bcb29b5a8ca1da875968085f22b7c0a50f29a4851396251de121c
```

this is actually a good advancement, we now have the value of :

- amount in the voucher
- the salt used for hashing
- Alice and Bob signatures

what we can do by these is : compute the same exact voucher hash that was used in the redeeming process, and since we know Alice's signature for this hash, we actually have a valid pair that verifies to Alice's public address

while i was doing this work, i thought it would work because in the permit function when they are not hashing, was an implementation typo, so they would only check if the hashed value of amount is used, but they are actually checking that in this line :

```
require(!usedHashes[bytes32(amount)], HashAlreadyUsed());
```

(yup i didn't read all the code carefully :/)

now my plan will not work!! because the idea was to cast the hash value we computed before into `uint256` and then use it as the amount, and they would check its hashed value only

well, hopefully there is a save for this!

lets take a look at `ECDSA.recover` to see how it works

```

function tryRecover(
    bytes32 hash,
    uint8 v,
    bytes32 r,
    bytes32 s
) internal pure returns (address recovered, RecoverError err, bytes32 errArg) {
    // EIP-2 still allows signature malleability for ecrecover(). Remove this possibility and make the signature
    // unique. Appendix F in the Ethereum Yellow paper (https://ethereum.github.io/yellowpaper/paper.pdf), defines
    // the valid range for s in (301):  $0 < s < \text{secp256k1n} \div 2 + 1$ , and for v in (302):  $v \in \{27, 28\}$ . Most
    // signatures from current libraries generate a unique signature with an s-value in the lower half order.
    //
    // If your library generates malleable signatures, such as s-values in the upper range, calculate a new s-value
    // with  $0\text{x}\text{FFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141} - s1$  and flip v from 27 to 28 or
    // vice versa. If your library also generates signatures with 0/1 for v instead 27/28, add 27 to v to accept
    // these malleable signatures as well.
    if (uint256(s) > 0x7FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF5D576E7357A4501DDFE92F46681B20A0) {
        return (address(0), RecoverError.InvalidSignatureS, s);
    }

    // If the signature is valid (and not malleable), return the signer address
    address signer = ecrecover(hash, v, r, s);
    if (signer == address(0)) {
        return (address(0), RecoverError.InvalidSignature, bytes32(0));
    }

    return (signer, RecoverError.NoError, bytes32(0));
}

```

without digging into all the code details, all we need to know is that this line is the one responsible for recovering the signer address (Alice's public address)

`ecrecover` is a built-in solidity function that implements the `ecdsa` verification algorithm, the curve used is `secp256k1` :

Standard curve database

GitHub

secp256k1

256-bit prime field Weierstrass curve.

A Koblitz curve.

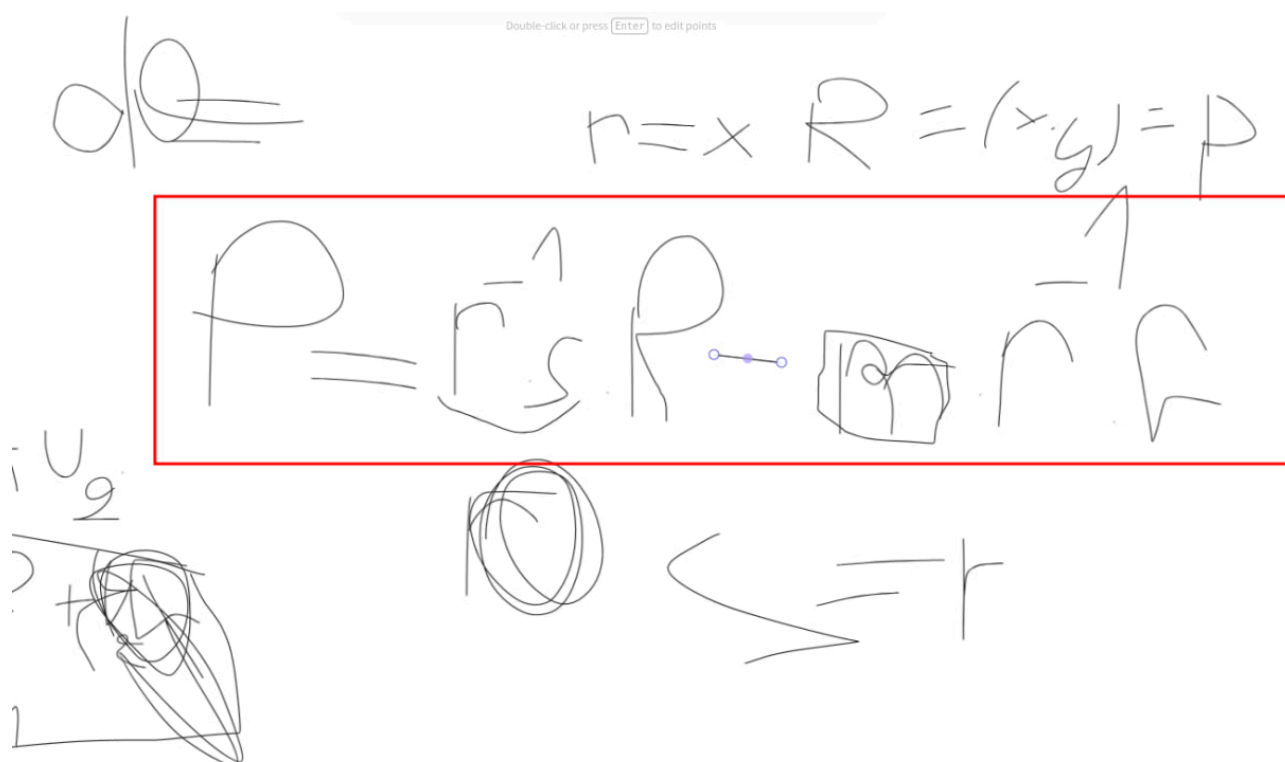
Also known as: **ansip256k1**

$$y^2 \equiv x^3 + ax + b$$

Parameters

Name	Value
p	0xfffefffffc2f
a	0x00
b	0x0007
G	(0x79be667ef9dcbbac55a06295ce870b07029bfcd2dce28d959f2815b16f81798, 0x483ada7726a3c4655da4fbfc0e1108a8fd17b448a68554199c47d08ffb10d4b8)
n	0xfffffffffffffffffffffffffffffffffebaaedce6af48a03bbfd25e8cd0364141
h	0x1

lets go one step deeper and look at how the recovery is done (this is my favorite part ^-^):



so this some of my writing while i was solving the challenge (sorry for the ugly drawing haha)

as we can see this is how the public point of the verifier is computed (and then from it they compute the address)

notice that i wrote $m \cdot r^{-1}$, generally it is $z \cdot r^{-1}$ where z is the hash message, but since they are not hashing in the challenge, i will use m

here we are doing $u_1 \cdot R + u_2 \cdot R$ right?! (u_1 and u_2 are just shortcuts)

the scalars u_1 and u_2 are mod n , that is because we are working in the elliptic curves group

more precisely $u_1 \cdot R$ is actually an abbreviation for $R + R + R \dots u_1$ times, and point addition is done modulo the order of the finite cyclic group of the elliptic curves

with this in mind, it is important to notice that this will open a way for collisions !!

because since we are not hashing, the amount we pass is directly used as m in the verification, and when working mod n , both m and $m + n$ are the same number, but they are different hashes/numbers in the require check

2- use the same hash + n and signature used in the redemption

now since we proved that m and $m+n$ will be equal inside `ECDSA.recover`, we can `uint256(hash) + n` instead of `uint256(hash)` (notice that by hash i mean the voucher hash for Alice they used in the redeem function) and it will pass the require check because it is actually a different value outside the verification

this should solve the challenge, right :) ?

well, i did hit a bad luck :// , $\text{hash} + n$ overflowed (>256 bits) and $\text{hash} - n$ is negative

the whole work didn't work at the end



3- i found another solution

remember the previous verification equation, we will use that.

lets forget about trying to use the same signature of Alice, and forge a different signature that recovers Alice's public address (public point)

this is actually possible because they are not hashing :)

assume we take $m = 0$ (this wouldn't be possible in case they were hashing because a hash can't be zero)

the equation will become : $P = u_1 * R$, where u_1 is $r^{-1} * s$

what would happen if we choose $r = P_x$ (the goal is to make $R = P$), and $s=r$, in this case $u_1 * R$ will be equal to P , where P is the public point of Alice, This is the solution for the challenge for sure !!

well for $m=0$, we will steal zero tokens, but we can take $m = n$ which is equivalent to 0

now there is something missing to fully craft this exploit, which is the x coordinate of P

we can't get it directly from Alice's address, that is just a part of the hash of P

recovering the x coordinate of Alice's public point

we stated before that we have a valid pair that verifies to Alice's public point, which is the signature and voucher hash

first we compute R

we are still missing the y value for R , we compute it like this :

```
p = 0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffefffffc2f
a = 0
b = 7
E = EllipticCurve(GF(p), [a,b])
r = 0xab1dcd2a2a1c697715a62eb6522b7999d04aa952ffa2619988737ee675d9494f
s = 0x2b50ecce40040bcb29b5a8ca1da875968085f22b7c0a50f29a4851396251de12
z = 0x87f1c8cd4c0e19511304b612a9b4996f8c2bd795796636bd25812cd5b0b6a973
n =
1157920892373161954235709850086879078528375642790749043826051631415181614943
37
r_inv = pow(r, -1, n)
G = E(0x79be667ef9dcbbac55a06295ce870b07029bfcdb2dce28d959f2815b16f81798,
0x483ada7726a3c4655da4fbfc0e1108a8fd17b448a68554199c47d08ffb10d4b8)
yy = pow(r, 3, p) + 7
```

```

y = modular_sqrt(yy,p)
if y%2 == 1 :
    R = E(r,y)

```

Alice's signature is (r,s,v) where v tells us which y we take (modular_sqrt returns 2)

v was 28 which means y should be odd (we take the even value if v=27)

after recovering R we compute P by : (python code)

```

P = r_inv * s * R - z * r_inv * G

```

and this is the solidity code i used to extract r,s,v from Alice's signature:

```

bytes memory signature =
hex"ab1dcd2a2a1c697715a62eb6522b7999d04aa952ffa2619988737ee675d9494f2b50ecce
40040bcb29b5a8ca1da875968085f22b7c0a50f29a4851396251de121c";

bytes32 r;
bytes32 s;
uint8 v;

assembly ("memory-safe") {
    r := mload(add(signature, 0x20))
    s := mload(add(signature, 0x40))
    v := byte(0, mload(add(signature, 0x60)))
}

console.logBytes32(r);
console.logBytes32(s);
console.log(v);

```

this is the x value we recovered :

```
23454111434303081435929071042469611746672141632665808216069109536628528697239
```

now we craft our signature by setting r = s = x and v = 27

there is an extra security alert that forbids values of $s > n/2$, in our case s was smaller, but if it weren't, can simply replace s by $n - s$ and set v = 28

```

bytes memory tokenOwnerSignature=
hex'33da8e7fe906411e4fc12842632ec77c2aee6a4324a4a3ca554b56667e4ccf9733da8e7f
e906411e4fc12842632ec77c2aee6a4324a4a3ca554b56667e4ccf971b';

uint256 amount =
1157920892373161954235709850086879078528375642790749043826051631415181614943

```

```
37;
```

```
    bytes32 permitAcceptHash =  
keccak256(abi.encodePacked(0xA11CE84AcB91Ac59B0A4E2945C9157eF3Ab17D4e,0x284C  
d6B2b7D4dC8a754c8fC34858529db5039eBb, amount));
```

these are the values we will be passing to the permit function, last thing is we compute the signature for permitAcceptHash using our private key, this is straight forward

we can check that they recover to Alice's public point by running :

```
address tokenOwner = ECDSA.recover(bytes32(amount), tokenOwnerSignature);
```

and print the address

after checking that we are correct, we execute :

```
instance.permit(amount,0x284Cd6B2b7D4dC8a754c8fC34858529db5039eBb,tokenOwnerSig  
nature,spenderSignature);
```

where 0x28... is the address i will transferring tokens to

and we solve the challenge by stealing the tokens using :

```
_approve(tokenOwner, spender, amount);
```

this is the full solidity code :

```
// SPDX-License-Identifier: MIT  
pragma solidity ^0.8.0;  
  
import "../src/EllipticToken.sol";  
import {Script} from "forge-std/Script.sol";  
import "../forge-std/console.sol";  
  
contract Solver is Script {  
    EllipticToken instance =  
    EllipticToken(0x54056d5f90Fa1352602167405A91c6071BaB9096);  
  
    function run() external {  
        vm.startBroadcast(vm.envUint("PRIVATE_KEY"));  
        address Alice = 0xA11CE84AcB91Ac59B0A4E2945C9157eF3Ab17D4e;  
        bytes32 salt =  
0x04a078de06d9d2ebd86ab2ae9c2b872b26e345d33f988d6d5d875f94e9c8ee1e;  
        bytes memory signature =  
hex"ab1dcd2a2a1c697715a62eb6522b7999d04aa952ffa2619988737ee675d9494f2b50ecce  
40040bcb29b5a8ca1da875968085f22b7c0a50f29a4851396251de121c";
```

```

    bytes32 r;
    bytes32 s;
    uint8 v;

assembly ("memory-safe") {
    r := mload(add(signature, 0x20))
    s := mload(add(signature, 0x40))
    v := byte(0, mload(add(signature, 0x60)))
}

    console.logBytes32(r);
    console.logBytes32(s);
    console.log(v);
    bytes32 voucherHash =
keccak256(abi.encodePacked(uint256(1000000000000000000000000), Alice, salt));
    uint256 voucherHashInt = uint256(voucherHash) ;
    console.logBytes32(voucherHash);


    bytes memory tokenOwnerSignature=
hex'33da8e7fe906411e4fc12842632ec77c2aee6a4324a4a3ca554b56667e4ccf9733da8e7f
e906411e4fc12842632ec77c2aee6a4324a4a3ca554b56667e4ccf971b';
    uint256 amount =
1157920892373161954235709850086879078528375642790749043826051631415181614943
37;
    bytes32 permitAcceptHash =
keccak256(abi.encodePacked(0xA11CE84AcB91Ac59B0A4E2945C9157eF3Ab17D4e,0x284C
d6B2b7D4dC8a754c8fC34858529db5039eBb, amount));

    console.logBytes32(permitAcceptHash);


    bytes memory spenderSignature =
hex"48cbfa3546c0aa10fae491e66c3c76e2aeb3db53943e77e0ceedcf449d90ec96455a1f41
43982b6aebd28805ca9055b277978ebabe0cd17b02754557d80260721b";


    address tokenOwner = ECDSA.recover(bytes32(amount),
tokenOwnerSignature);
    console.log(tokenOwner);


instance.permit(amount,0x284Cd6B2b7D4dC8a754c8fC34858529db5039eBb,tokenOwner
Signature,spenderSignature);


instance.transferFrom(Alice,0x284Cd6B2b7D4dC8a754c8fC34858529db5039eBb,uint2
56(1000000000000000000000000));

```

```
}  
}
```

****resources :**

for elliptic curves, i would recommend :

<https://rareskills.io/zk-book> module 1, chapter 4 to 9, it has one of the best explanations i found about groups and elliptic curves, looking at elliptic curves as a group and finite field is actually really helpful

https://github.com/elikaski/ECC_Attacks this repo is also good, i like the explanation

<https://cryptohack.org/challenges/ecc/> this challenges can help you understand and apply more

<https://hackropole.fr/fr/crypto/> you can find more difficult challenges here

THANK YOU !!! IF YOU READ THIS I REALLY REALLY APPRECIATE YOUR TIME